

# JORDAN B. S. VAN BEECK

Asteroseismology | Stellar Astrophysics | FOSS



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## SCIENTIFIC EDUCATION

Ph.D., Astronomy & Astrophysics, KU Leuven (Belgium), September 2023

*Asteroseismology of Kepler B stars: internal magnetism and non-linear mode coupling*

Promoters: Prof. Dr. C. Aerts, Prof. Dr. T. Van Hoolst, Prof. Dr. D. M. Bowman

M.Sc., Astronomy & Astrophysics, KU Leuven (Belgium), July 2019 – *Magna Cum Laude*

*The influence of an interior magnetic field on gravity-mode oscillations of intermediate-mass stars*

Promoters: Prof. Dr. C. Aerts, Dr. T. Van Reeth, Prof. Dr. D. M. Bowman

M.Sc., Chemistry, University of Antwerp (Belgium), September 2017 – *Magna Cum Laude*

*Characterization of radioactive particles*

Promoters: Prof. Dr. K. Janssens, Prof. Dr. B. Salbu, Prof. Dr. O.-C. Lind, M.Sc. G. Nuyts

B.Sc., Chemistry, University of Antwerp (Belgium), September 2015 – *Magna Cum Laude*

*Atomic scale reactive MD studies of DNA oxidation for plasma oncology: the role of  $H_2O_2$  and  $HO_2$*

Promoters: Prof. Dr. A. Bogaerts, Prof. Dr. E. Neyts, Dr. C. Verlackt

## EMPLOYMENT

**Heidelberg Institute for Theoretical Studies (HITS)**, TOS group, Heidelberg, Germany

Postdoctoral Researcher Asteroseismology February 2025 – March 2027

Project: higher-order rotational effects on mixed-mode oscillations of red giants

**KU Leuven**, Department of Physics and Astronomy, Institute of Astronomy, Leuven, Belgium

Research Associate, Asteroseismology, September 2023 – December 2023

Project: non-linear asteroseismic modeling of slowly pulsating B stars

## AWARDS AND FELLOWSHIPS

|             |                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021 - 2022 | FWO(Research Foundation Flanders) long research stay grant (€12 738) for a stay as a <a href="#">VSR</a> at TAPIR, Caltech, CA, USA with Prof. Dr. J. Fuller. |
| 2019 - 2023 | 4-year PhD Fellowship, Department of Physics and Astronomy, KU Leuven                                                                                         |
| 2019        | Paul Smeyers Prize, KU Leuven<br>Awarded to the annual best master's thesis in Astronomy & Astrophysics at the June examination session.                      |

## TEACHING EXPERIENCE

**Lecturer, Heidelberg University**

|      |                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Theoretical introduction for 'Asteroseismology'<br>(Master-level physics course that introduces various aspects of asteroseismology.) |
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**Teaching Assistant, Department of Physics and Astronomy, KU Leuven**

|             |                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 - 2023 | K01B4A: 'Natuurkunde met elementen van wiskunde' I (taught in Dutch)<br>(Introductory physics course for the first-year students in the B.Sc. of pharmaceutical sciences)                                                                                            |
| 2019 - 2021 | K01B4A B: 'Natuurkunde met elementen van wiskunde' I II (taught in Dutch)<br>(Introductory physics courses for the first-year students in the B.Sc. of pharmaceutical sciences)<br>English translation of course titles: 'Physics with elements of mathematics' I II |

## VISITING RESEARCH POSITIONS (ABRIDGED)

|                   |                                                                                                                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11/2021 - 06/2022 | <p>Visiting Student Researcher (VSR) at TAPIR, Caltech, CA, USA</p> <p>Project goal: development of computational tools that aid in explaining oscillation mode amplitude limitation in slowly pulsating B stars</p> <p>Mentor: Prof. Dr. J. Fuller <i>Funding: FWO long research stay grant</i></p> |
| 11/2021 - 12/2021 | <p>Transport in Stellar Interiors Program Affiliate (<a href="#">workshop</a>, <a href="#">conference</a>) at KITP, UC Santa Barbara, CA, USA</p> <p>Mentor: Prof. Dr. L. Bildsten <i>Funding: FWO long research stay grant</i></p>                                                                  |

## SEMINARS/CONFERENCE PRESENTATIONS (ABRIDGED)

|               |                                                                                                                                                                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 2026     | <p><b>POSTER PRESENTATION</b> (<a href="https://doi.org/10.5281/zenodo.21083361">doi:10.5281/zenodo.21083361</a>)</p> <p>TASC10 / KASC17 workshop/conference at Aarhus University, Denmark.</p>                                                                                   |
| July 2025     | <p><b>CONTRIBUTED TALK</b> (<a href="https://doi.org/10.5281/zenodo.17241657">doi:10.5281/zenodo.17241657</a>)</p> <p>TASC9 / KASC16 workshop/conference at ISTA, Austria.</p>                                                                                                    |
| March 2023    | <p><b>GOOD VIBRATIONS SEMINAR</b> (<a href="https://youtu.be/dWaEkri_s2I?feature=shared">https://youtu.be/dWaEkri_s2I?feature=shared</a>)</p> <p>The Good Vibrations seminar series, S3E8.</p>                                                                                    |
| February 2023 | <p><b>AD-VALVAS SEMINAR</b> (<a href="https://youtu.be/nTWZoLCMa_g?feature=shared">https://youtu.be/nTWZoLCMa_g?feature=shared</a>)</p> <p>Ad-valvas seminar of the Institute of Astronomy, KU Leuven, Belgium.</p>                                                               |
| July 2022     | <p><b>POSTER PRESENTATION</b> (additional info: <a href="https://doi.org/10.5281/zenodo.6814504">doi:10.5281/zenodo.6814504</a>)</p> <p>TASC6 / KASC13 workshop/conference at KU Leuven, Belgium.</p>                                                                             |
| November 2021 | <p><b>CONTRIBUTED TALK</b> (<a href="https://doi.org/10.26081/K6VH15">doi:10.26081/K6VH15</a>)</p> <p>Transfer in Stellar Interiors conference (part of the Probes of Transport in Stars program), Kavli Institute for Theoretical Physics (KITP), UC Santa Barbara, CA, USA.</p> |

## PROFESSIONAL ACTIVITIES

### Conference organization

|           |                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------|
| July 2022 | <p>Member of the Local Organizing Committee</p> <p>TASC6 / KASC13 workshop/conference in Leuven, Belgium</p> |
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### Professionally associated as

|             |                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------|
| 2026 - ...  | Junior Member of the <i>International Astronomical Union (IAU)</i>                                    |
| 2022        | Fellow (Graduate Student) of the <i>Royal Astronomical Society (RAS)</i>                              |
| 2020 - 2022 | Graduate Student Member of the <i>American Astronomical Society (AAS)</i>                             |
| 2020 - 2022 | Graduate Student Member of the <i>Royal Netherlands Astronomical Society (KNA)</i>                    |
| 2019 - 2023 | Graduate Student Member of the <i>International Research Network for Nuclear Astrophysics (IReNA)</i> |

## DIGITAL PROFICIENCY

